



Department of Education

**STEM Education
National Schools of Excellence
Teacher's Guide
Grade 12
Term 2**

<u>DOMAIN</u>	<u>Phylum</u>
Science	Physics
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

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ISBN

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SECRETARY'S MESSAGE

This Physics Syllabus is to be used exclusively by teachers of Physics to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Physics Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Physics to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD

Secretary for Education

INTRODUCTION

Physics is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Physics help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics.

The Grade 12 Physics Syllabus entails ten (6) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Physics is a highly specialized subject students undertaking Physics are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher Guide are sequenced to guide teachers of Physics through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Biophysics	1. Introduction to Biophysics 2. Genomics – DNA Sequencing 3. Proteomics
	12.2 Bioelectronics	1. Applications of Bioelectronics
HEALTH AND MEDICINE	12.1 Nuclear Medicine	1. Introduction to Nuclear Medicine 2. Nuclear Medicine - Diagnosis 3. Nuclear Medicine - Therapy
ENERGY	12.1 Nuclear Energy	1. Introduction to Nuclear Energy 2. Nuclear Reactions – Fission and Fusion 3. Nuclear Fission Reactors
BATTERY TECHNOLOGY	12.1 Inductive Charging	1. Electromagnetism 2. Induction Charging 3. Hybrid Electric Vehicles

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Physics Bridging Syllabus students were introduced to the basic concepts and knowledge of Physics in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Physics as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM PHYSICS CONTENT OVERVIEW – GRADE 12, TERM 2

Table 1.1 Sequencing & Benchmarking STEM Physics Content Standards

Setting STEM Physics Content Standards for Progressive Learning & Assessment			
Thema	Energy	Broad Content Standard	
Module 12.1	Nuclear Energy	12.1: Conceptualize physics of nuclear energy and use it to design a simple power plant	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1: Introduction to Nuclear Energy	12.1.1a: Clearly define and explain nuclear energy. 12.1.1b: Eloquentlly describe the fundamental particles that constitute to create the atomic structure. 12.1.1c: Define isotopes and describe how the nucleus composition determines whether it is a stable or unstable isotope (radioisotope). 12.1.1d: Define and explain radioactivity and radioactive decay. 12.1.1e: Recognize and distinguish the different types of radioactive decay. 12.1.1f: Explain the law of radioactive decay and half-life and describe why there are different rates of radioactive decay for different radioisotopes. 12.1.1g: Describe Understand the concept of radioactive series.	Students' ability to master the knowledge and skills of; • Atomic structure and fundamental particles • Radiation and Radioactive decay • Nuclear fission and fusion reactions • Nuclear Reactors and general nuclear power plant setup	Research Project + Written Test
2: Nuclear Reactions – Fission and Fusion	12.1.2a: Clearly explain binding energy and how it can determine which substances will undergo nuclear fission and fusion. 12.1.2b: Clearly differentiate nuclear fission and nuclear		

	fusion and list some of the materials used. 12.1.2c: Describe the concept of mass-energy equivalence in a nuclear reaction. 12.1.2d: Calculate the energy released during nuclear fission.		
3: Nuclear Fission Reactors	12.1.3a: Briefly describe the different stages involved in processing uranium for use in fission reactors. 12.1.3b: Define and generally explain how a nuclear power plant works. 12.1.3c: List and explain the different components of a power plant with their operations. 12.1.3d: Differentiate and describe how a boiling water and pressurized water power plant works. 12.1.3e: Describe the efficiency of a nuclear power plant.		
Thema	Battery Technology	Broad Content Standard	
Module 12.1	Inductive Charging	12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.	
Topics	Performance Indicators	Module Benchmark Students' ability to master the knowledge and skills of;	Assessment Tasks
1: Electromagnetism	12.1.1a: Explain the phenomena of electromagnetism. 12.1.1b: Describe how a magnetic field generates current and vice versa. 12.1.1c: Clearly explain how electromagnets and	- The basic working mechanism of Fossil Powered Vehicles. - The basic working mechanism of Electric Vehicles.	Research Project + Written Test

	transformers utilize electromagnetic induction.	The basic working mechanism of Hybrid Electric Vehicles.	
2: Inductive Charging	12.1.2a: Describe the process involved in Inductive Charging 12.1.2b: How inductive charging is utilized in charging electric vehicles		
3: Hybrid Electric Vehicles	12.1.3a: Differentiate between IC engines and Hybrid Electric Vehicles. 12.1.3b: Identify and distinguish between the different types of Hybrid Electric Vehicles. 12.1.3c: Explicitly explain and describe how hybrid electric vehicles integrate battery technology. 12.1.3d: Differentiate between IC engines and hybrid Electric Vehicles. 12.1.3e: Describe the role of batteries and inductive charging for the future electric vehicle market.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	12.1	12.1
Energy/ Module 12.1: Nuclear Energy	12.1 Conceptualize the knowledge physics in the atomic structure that contributes to harnessing nuclear energy 12.2 Grasp the concept of nuclear energy in and its applications in large scale power generation in fission and fusion power plants.	✓	
Battery Technology/ Module 12.1: Inductive Charging	12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.		✓

THEMA: ENERGY

MODULE 12.1: NUCLEAR ENERGY

4 weeks

Mentioned previously, there are different types of energy – thermal, electrical, solar, hydro, mechanical, chemical, biological, etc. In this module, we will focus on nuclear energy.

Nuclear energy is the energy stored in an atom's nucleus. Every object in the universe is made up of atoms, which are small particles. The bonds that hold these atoms together contain a tremendous amount of energy which can be used to generate electricity. However, the energy must first be released. Nuclear fusion and nuclear fission are two methods for releasing nuclear energy. Nuclear fusion releases energy when atoms unite or fuse together to form a bigger atom and the sun generates energy in this manner. Nuclear fission releases energy when atoms are torn apart to form smaller atoms and it is the most common method for harnessing nuclear energy in which nuclear fission power facilities generate electricity.

In this module, students will learn about nuclear energy, the materials involved in the process of harnessing nuclear energy through fission reaction, the applications of nuclear energy (in terms of nuclear technologies), and specifically look at how nuclear power plants generate electricity.

Prerequisites:

- Grade 11 Term 3, STEM Physics (Thema: ENERGY)
- Grade 11 Term 3, STEM Engineering (Thema: ENERGY)

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Conceptualize the knowledge physics in the atomic structure that contributes to harnessing nuclear energy

12.2 Grasp the concept of nuclear energy in and its applications in large scale power generation in fission and fusion power plants.

Content Benchmark

Students' ability to master the knowledge and skills of;

- Atomic structure and fundamental particles
- Radiation and Radioactive decay
- Nuclear fission and fusion reactions
- Nuclear Reactors and general nuclear power plant setup

Topic 1: Introduction to Nuclear Energy

Performance Indicators

Students are expected to;

12.1.1a: Clearly define and explain nuclear energy.

12.1.1b: Eloquently describe the fundamental particles that constitute to create the atomic structure.

12.1.1c: Define isotopes and describe how the nucleus composition determines whether it is a stable or unstable isotope (radioisotope).

12.1.1d: Define and explain radioactivity and radioactive decay.

12.1.1e: Recognize and distinguish the different types of radioactive decay.

12.1.1f: Explain the law of radioactive decay and half-life and describe why there are different rates of radioactive decay for different radioisotopes.

12.1.1g: Describe Understand the concept of radioactive series.

Review Exercise 1

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Review Exercise 2

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Review Exercise 3

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Topic 2: Nuclear Reactions - Fission and Fusion

Performance Indicators

Students are expected to;

12.1.2a: Clearly explain binding energy and how it can determine which substances will undergo nuclear fission and fusion.

12.1.2b: Clearly differentiate nuclear fission and nuclear fusion and list some of the materials used.

12.1.2c: Describe the concept of mass-energy equivalence in a nuclear reaction.

12.1.2d: Calculate the energy released during nuclear fission.

Review Exercise 1

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Topic 3: Nuclear Fission Reactors

Performance Indicators

Students are expected to;

12.1.3a: Briefly describe the different stages involved in processing uranium for use in fission reactors.

12.1.3b: Define and generally explain how a nuclear power plant works.

12.1.3c: List and explain the different components of a power plant with their operations.

12.1.3d: Differentiate and describe how a boiling water and pressurized water power plant works.

12.1.3e: Describe the efficiency of a nuclear power plant.

Assignment

Prepare the Assignment and hand to students to attempt

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Nuclear Energy	12.1 Conceptualize the knowledge physics in the atomic structure that contributes to harnessing nuclear energy 12.2 Grasp the concept of nuclear energy in and its applications in large scale power generation in fission and fusion power plants.	1. Introduction to Nuclear Energy	Assignment + Written Quiz/Test
		2. Nuclear Reactions – Fission and Fusion	
		3. Nuclear Fission Reactors	

Sample Assessment Marking Guide

Sample Assessment Marking Guide			
Topic	Assessment Task	Assessment Criteria	Total Marks
1. Introduction to Nuclear Medicine	Written Quiz	What is nuclear energy?	2
		What are fundamental particles? List and explain the two types	2
		What are the four main forces in the universe? Describe how they relate to the atomic structure of matter.	4
		What is radioactive decay? List and differentiate them. Also include the generic equations for each.	4
		What is antimatter?	1
		Describe how the lepton number is conserved in Beta plus decay and Electron Capture.	4
		What is the difference between Beta-Minus and Beta Plus decay?	3
Subtotal			20
Module 12.1 (All three Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	15
		Correctly answers all multiple choice	10
		Correctly answers short answer questions	30
		Subtotal	55
Module TOTAL			75

THEMA: BATTERY TECHNOLOGY

Introduction

A battery is a storage device and converter for chemical energy and electrical energy. It has one or more electrochemical cells that are connected to external circuits. The resulting conversion of the stored chemical energy to electrical energy powers the electrical devices such as flashlights, electric cars and mobile phones. When looking at the battery, the positive terminal is known as the cathode while the negative terminal is known anode.

Electrons flow from the anode then to the external circuit and onto the cathode. When the battery is connected in a circuit to give electricity, there is a reaction where electrons are gained and lost and this reaction is known as the reduction-oxidation reaction or in short, the redox reaction. The redox reaction in the battery converts the stored high chemical energy which reacts to low energy products and the free energy difference between the high energy of the reactants and the low energy of the products results as electrical energy to the external circuit.

Listed below are few of the many career pathways for students who are interested in the field of battery technology development.

- Cell Developer
- Process Engineering Technician
- Mechanical Engineer (Battery Performance Testing)
- Battery Product Design Engineer – Cell Mechanical Characterization
- Electrode Coating Process Chemistry/Engineer
- Battery Cell Engineer, and etc.

MODULE 12.1: INDUCTIVE CHARGING

4 weeks

Also known as wireless charging, inductive charging is a method of charging the battery in portable electrical devices without having to plug the device directly into a power socket. In most cases, smartphones were one of the earlier devices that were capable of utilizing this electromagnetic phenomenon.

In recent years, inductive charging is now one of the alternate modes for charging electric vehicles. In this module, we will discuss the concept of electromagnetism and how inductive charging is used to “refuel” electric vehicles.

Prerequisites:

- Grade 11 Term 3, STEM Engineering (Thema: BATTERY TECHNOLOGY)

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.

Content Benchmark

Students’ ability to master knowledge and skills of;

- Electromagnetism
- Inductive Charging
- Basic mechanism of a Hybrid Electric Vehicle

Content Benchmark

Students’ ability to master the knowledge and skills of;

- The basic working mechanism of Fossil Powered Vehicles.
- The basic working mechanism of Electric Vehicles.
- The basic working mechanism of Hybrid Electric Vehicles.

Topic 1: Electromagnetism

Performance Indicators

Students are expected to;

12.1.1a: Explain the phenomena of electromagnetism.

12.1.1b: Describe how a magnetic field generates current and vice versa.

12.1.1c: Clearly explain how electromagnets and transformers utilize electromagnetic induction.

Practical Assignment / Experiment 1

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

Title: Magnetic Field around a Current-carrying Conductor

Introduction

The discovery of this relationship between magnetism and electricity was, like so many other scientific discoveries, stumbled upon almost by accident. The Danish physicist Hans Christian Oersted was lecturing one day in 1820 on the possibility of electricity and magnetism being related to one another, and in the process demonstrated it conclusively by experiment in front of his whole class. By passing an electric current through a metal wire suspended above a magnetic compass, Oersted was able to produce a definite motion of the compass needle in response to the current. What began as a guess at the start of the class session was confirmed as fact at the end. Needless to say, Oersted had to revise his lecture notes for future classes. His discovery paved the way for a whole new branch of science – electromagnetism.

In this experiment, we will perform a basic investigation on the magnetic field around a current carrying conductor.

Apparatus:

- 1x 9V Battery with holder
- 2x Hookup wires with alligator clips
- compass
- Stop watch

Method:

1. Connect your wires to the battery leaving one end unconnected so that the circuit is not closed.
2. One student should be in charge of limiting the current flow to 10 seconds. This is to preserve battery life as well as to prevent overheating of wires and battery contacts.
3. Place the compass close to the wire.
4. Close the circuit and observe what happens to the compass
5. Reverse the polarity of the battery and close the circuit. Observe what happens to the compass.

Conclusions:

Use your observations to answer the following questions:

1. Does a current flowing in a wire generate a magnetic field?
2. Is the magnetic field present when the current is not flowing?
3. Does the direction of the magnetic field produced by a current in a wire depend on the direction of the current flow?
4. How does the direction of the current affect the magnetic field?

(Refer to TRB)

Assignment / Research Project

Prepare the Assignment and hand to students to attempt

Resources: NIL

(Refer to TRB)

Topic 2: Induction Charging

Performance Indicators

Students are expected to;

12.1.2a: Describe the process involved in Inductive Charging

12.1.2b: How inductive charging is utilized in charging electric vehicles

Practical Assignment / Experiment 1

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

Title: Tesla Coil

Introduction:

Developed in 1891 by Nikola Tesla, the Tesla coil was created to perform experiments in creating high-voltage electrical discharges. It consists of a power supply, a capacitor and coil transformer set so that voltage peaks alternate between the two, and electrodes set so that sparks jump between them through the air. Used in applications from particle accelerators to televisions and toys, a Tesla coil can be made from electronics store equipment or from surplus materials.

In this experiment, we will create a Tesla Coil and do a basic test that displays the electromagnetic phenomena of wireless power transfer.

CAUTION: Do not use near persons with pacemakers, sensitive electronics or flammable materials.

NOTE: To avoid overheating, only power your Tesla Coil for short durations of no more than 20 seconds or less.

Apparatus:



- Breadboard Circuit (A-J/1-17)
- MJE3055T Transistor with Heat Sink
- 3x 104 .1uF Ceramic Capacitors
- 1K Resistor
- Solid Core 16 ga. Insulated Copper Wire, ~1.5ft.
- PVC Pipe 2" x 2.5" diam.
- AWG 27 Insulated Magnet Wire
- PVC Pipe 7" x 2" diam.
- 3" Steel Washer
- 5x Jumper Wires
- 12v/1A Power Supply
- 2x 8" x 10" Plexiglass Sheets
- 4x 5/15" Threaded Rod
- 16x 5/16" Nuts
- 16x 5/16" Washers
- 8x 5/16" Rubber End Caps

Methodology:

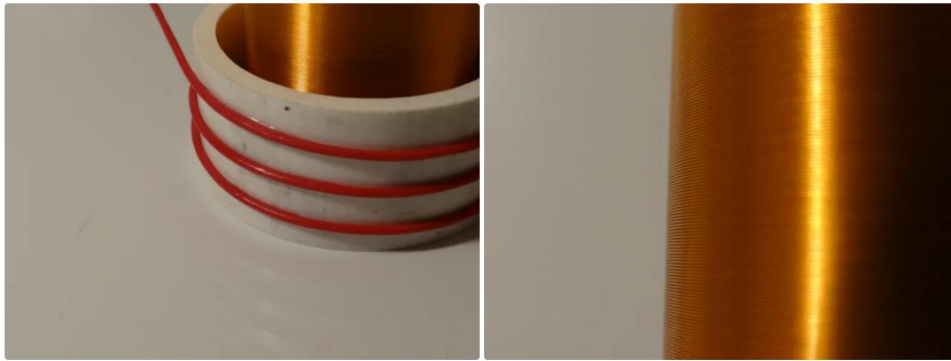
In this experiment, we will create three separate components:

- I. **Primary Circuit**
- II. **Secondary Circuit**
- III. **Enclosure**

I. **Primary Circuit**

The primary circuit will be responsible for generating the magnetic field when connected to the power supply.

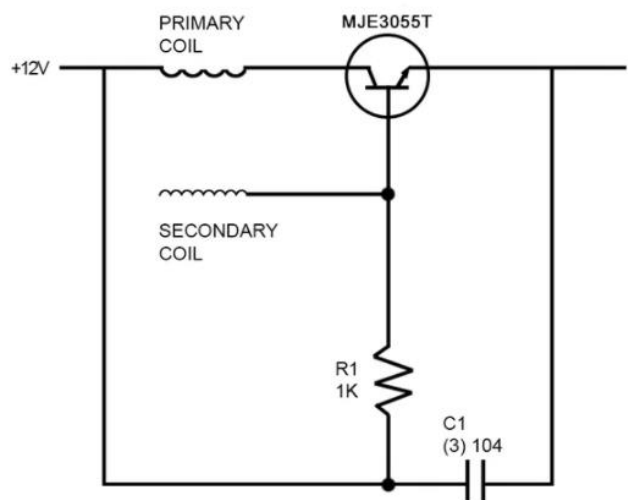
1. Firstly, we need to wind out coils accurately and precisely for them to function properly in the experiment. We begin with making the primary coil. We will wrap our short 2.5" PVC pipe with the 16 ga. Insulated Copper Wire making three rotations evenly spaced about 1/4" apart and secure with tape. Then strip the ends.



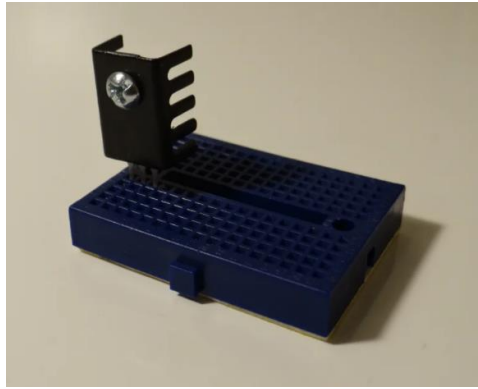
2. Next, take the 2" PVC and line up the magnet wire across about 1/4" from the bottom and secure it with tape leaving several inches extra on the end.
3. Then, wrap the magnet wire around several hundred times until we reach around 1/4" from the top. Be sure to wrap tightly, straight and without gaps between windings. Also, be sure to add a piece of tape every inch or so to keep everything secure. Once the top is reached, leave a couple inches of additional wire, cut and strip both ends by lightly, sand the ends of the wire. To firmly secure the windings, a tape can be used through each winding from top to bottom.
4. Next, press the stripped wire end between the top of the PVC and your 3" washer and secure with glue. This will act as your secondary coil and transmitter cap.

II. Secondary Circuit

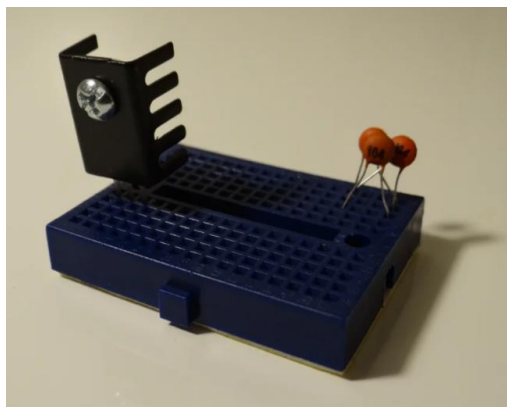
We will then construct a secondary circuit that combines to form the complete circuit. This will look similar to the diagram given below.



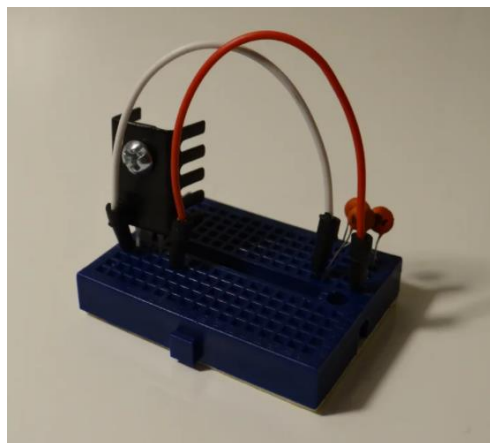
1. Firstly, install the three legs of the transistor in breadboard slots E1, E2 and E3 with the heat sink and front of the transistor facing back toward slot F.



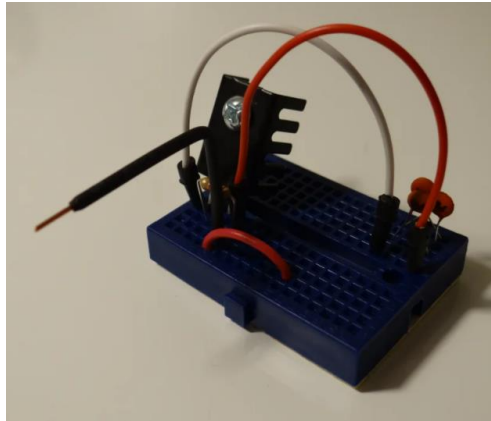
2. Secondly, insert the three capacitors into slots H14/H17, I14/I17 and J14/J17 respectively so that they are in parallel.



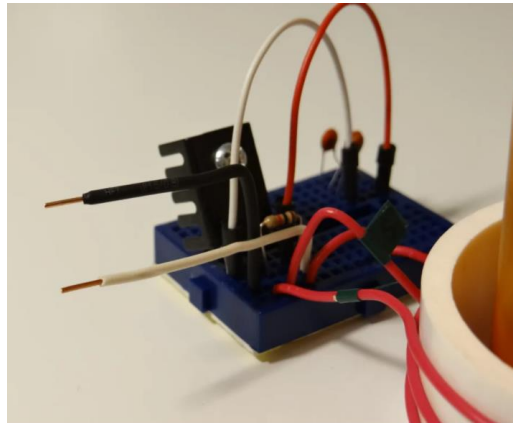
3. After this, connect the first leg of the transistor to one side of the capacitors with a jumper wire. Connect one end of a jumper wire to slot D1 and the other to F14.



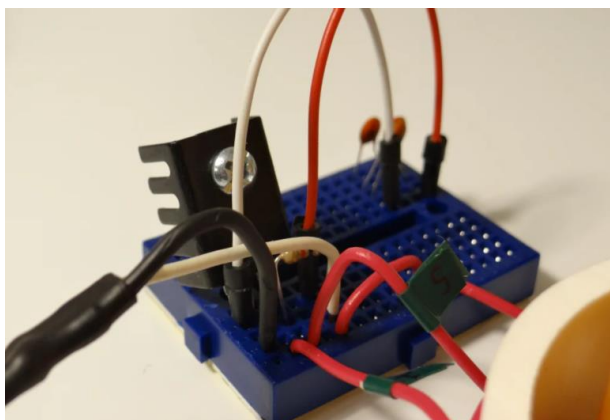
4. Next, connect a jumper wire from the other side of the capacitors back to where the ground will be. Connect one end of a jumper wire to slot F17 and the other end to slot D5.



5. Using the resistor, insert one of its ends on the same column, slot C5, and the other end to the end to the base of the transistor in slot C3.



6. Next, connect one last jumper wire to slot A5 and the other end to slot B11. This will allow for connection to the primary coil.
7. After connection, insert the constructed secondary coil into the primary coil, ensuring that it is kept at the centre.



8. The bottom wire of your primary coil can be inserted into slot A11. The top wire from your primary can be connected to slot A2. Connect your secondary coil by inserting the bottom wire into slot A3, and the base of your transistor.
9. Perform a check inspection of all connections before proceeding.
10. Lastly, connect the positive from your power supply (+) to slot B5, and connect the negative from your power supply (-) to slot B1.
11. You may now carefully test your circuit by plugging it in momentarily.

III. Enclosure

The enclosure will be constructed to serve four main functions:

- A display for our Tesla Coil.
 - To isolate the coil from flammable materials and sensitive electronics.
 - Stabilize and keep the coil upright and to
 - provide a platform for experimentation.
1. To construct the enclosure, we firstly, place a washer, nut and end cap on each of our threaded rods. Then we can drill a 5/16" hole in each corner of our plexiglass sheets.
 2. Then insert the four rods into the holes in one of your plexiglass sheets and add a washer and nut to secure, creating the base of the enclosure.
 3. Next, place your circuit and coil on top of the sheet, making sure it is centered, and remove the adhesive backing from the breadboard in order to affix it to the platform.
 4. Lastly, add a nut and washer to each of the rods, place the second plexiglass sheet on top and adjust in order to tightly hold the coil in place. Once secure, add an additional washer and nut to each rod, tighten and add an end cap to each.
 5. The Tesla Coil is now ready to be used for experimentation.



Experimentation and Results

1. Connect the power and watch as the florescent bulb lights up once placed near the coil. Using a digital multi-meter and the table below, observe the high voltage field at varied distances from your coil.

Distances (cm)	Observations
2	
4	
6	
8	
10	
12	
14	

Based on the experiment, write a lab report that records all your findings and analysis of the experiment.

Modified from Source: [Easy Tesla Coil! : 6 Steps \(with Pictures\) - Instructables](#)

(Refer to TRB)

Case Study / Class Discussion

Instruct the students to refer to their SRB to read on the case study and allow them to attempt the questions given. Prepare the class for a discussion based on their answers.

Resources: NIL

(Refer to TRB)

Topic 3: Hybrid Electric Vehicles

Performance Indicators

Students are expected to;

12.1.3a: Differentiate between IC engines and Hybrid Electric Vehicles.

12.1.3b: Identify and distinguish between the different types of Hybrid Electric Vehicles.

12.1.3c: Explicitly explain and describe how hybrid electric vehicles integrate battery technology.

12.1.3d: Differentiate between IC engines and hybrid Electric Vehicles.

12.1.3e: Describe the role of batteries and inductive charging for the future electric vehicle market.

Case Study / Class Discussion

Instruct the students to refer to their SRB to read on the case study and allow them to attempt the questions given. Prepare the class for a discussion based on their answers.

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Inductive Charging	12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.	1. Electromagnetism	Assignment + Written Quiz + Written Test
		2. Induction Charging	
		3. Hybrid Electric Vehicles	

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1. Electromagnetism	Written Quiz	What is meant by “nuclear” in nuclear medicine?	2
		What is medical Imaging and its relation to nuclear medicine?	2
		What are radiopharmaceuticals? List at least three different examples	4
		List at least one advantage and disadvantage of using nuclear medicine.	2
Subtotal			10
Module 12.1 (All three Topics)	Assignment (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	20
		Correctly answers all multiple choice	20
		Correctly answers short answer questions	10
		Subtotal	50
Module TOTAL			60

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules	Total Marks	% Weight of Modules
	6		
Theory Knowledge & skills – Test	Test (50)	/50	20
Application of knowledge – Assignments and projects	Assign.1 (30)	/30	30
End of Term Exam	(25 marks)	/25	50
Total marks for term one			100 marks

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Name s					Assessment Task				Total Score (/300)	Percentage Weight (%)				Achie vemen t Level [VHA, HA, SA,LA]
		Tests		Assignments				Project	Exam (/50)		T (20%)	A + P (30%)	Exam (50%)	Ttl CA	
1	Bill Joe	45	47	20	21	22	23	55	45	278	18.4	25.88	45	89.2 8	VHA
2	Cara Den	20	25	15	16	17	18	25	30	166	9	18.22	30	57.2 2	SA
3	Lea May	9	15	5	6	7	8	30	25	105	4.8	9.10	25	38.9	LA

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